

Empowering Through Education: Analyzing the Impact of the Right to Education Act on Intimate Partner Violence in India

Anas Khan*

Abstract

This paper exploits compulsory education reform in India to examine the effect of education on spousal violence. In 2009, the Government of India passed the Right to Education Act (RTE). The key feature of the reform was that students were required to finish 8 years of primary education till they reach 14 years of age. An instrumental variable difference-in-difference approach is employed that measures the exogenous change in education in different states across birth cohorts measured using birth month-year. The Act led to decrease in self-reported sexual and emotional violence, without changes in physical violence, or women's attitudes towards such violence. Further, the paper examined the potential mechanism through which increased education affects partner violence. The findings suggest changes in partner characteristics for example change in age and schooling gap and also better access to information contribute to reduced violence whereas I

*Doctoral Student, Indira Gandhi Institute of Development Research, Mumbai, 400065, India. Email: anas@igidr.ac.in

find no effect of other channels such as labor market outcomes, bargaining power and gender attitudes.

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JEL Classification: I21, I28, J12, J16, J24, K42.

1 Introduction

Spousal violence is a severe public health issue that disproportionately affects women. One in four women aged 15-49 years has been subjected to some form of violence from their intimate partner during their lifetime (World Health Organization, 2021¹). Such violence is more prevalent in developing countries, particularly where women have weak bargaining power within the household (WHO, 2013²). Women in violent relationships are more likely to report several health problems ranging from mental, physical, and reproductive health - which directly or indirectly affects women's well-being, labor market outcomes, and their children's outcomes (Campbell, 2002; Aizer, 2010). Given the high prevalence of domestic violence and its economic consequences, economists have contributed to the investigation of the cause and consequences of domestic violence.

There are several factors that contribute to the persistence of IPV, including societal attitudes and norms (González and Rodríguez-Planas, 2020), a lack of enforcement of existing laws and policies (Spohn and Tellis, 2012), and barriers faced by victims in seeking help and accessing services (Fanslow and Robinson, 2010; Fugate

¹<https://www.who.int/news-room/fact-sheets/detail/violence-against-women>

²https://apps.who.int/iris/bitstream/handle/10665/85239/9789241564625_eng.pdf

et al., 2005). The economic empowerment of women is one critical factor that affects partner violence. There are two conflicting hypotheses regarding the relationship between the empowerment of women and domestic violence. One strand claims that empowerment is a protective factor against IPV, as women are empowered through employment opportunities, conditional cash transfers, or access to welfare services, their outside options will improve, increasing bargaining status in their relationship and thus decreasing their exposure to violence (Farmer and Tiefenthaler, 2003; Aizer, 2010; Hidrobo and Fernald, 2013). Other side argues that better social and economic position of women encourages their partners to use violence as an instrument to extract resources (Bloch and Rao, 2002; Agarwal and Panda, 2007; Erten and Keskin, 2018). Additionally, women's better economic conditions may threaten their husband's social status so they may use violence to reassure their dominance, commonly known as the 'male backlash' effect (Eswaran and Malhotra, 2011; Macmillan and Gartner, 1999)

In this paper, I investigate the effect of a specific empowerment channel - Education - on the prevalence of intimate partner violence in India. Most of the Evidence on partner violence comes from income and employment opportunities, as a means of empowerment. Though Education is considered to be one of the most important factors that leads to women's empowerment, research on the impact of education on marriage violence outcomes is scarce. To the best of my knowledge, Akyol and Kirdar (2020) and Erten and Keskin (2018) for Turkey, and Zhou et al. (2021) for China have only been able to establish a causal link between education and partner violence. In the context of India, Eswaran and Malhotra (2011) has shown a negative correlation between women's education and domestic violence but failed to establish a causal re-

lationship. This study aims to fill this gap by studying the effect of schooling reform in India on the incidence of intimate partner violence.

Indian government has implemented several laws and policies to address Intimate partner violence that includes Protection of Women from Domestic Violence Act 2005³ and the Criminal Law (Amendment) Act 2013⁴. Despite these efforts, the problem persists and IPV continues to affect women significantly due to the lack of enforcement of existing laws and policies. As per NFHS-4 conducted in 2015-16, 33% of ever-married women aged 15-49 have experienced some form of partner violence. The same NFHS survey reports a sharp decline in all forms of spousal violence with Education, only 1 in 5 women with 12 or more years of schooling report ever facing partner violence. This association between education and marriage violence warrants the importance of examining education as an alternative tool to address the issue of partner violence. In societies such as India, women are more likely to experience violence due to the prevalence of traditional gender roles. Education can help women by changing their views about traditional gender norms and help them to take control of their life. Besides, Education may also lead women to make better decisions about marriage and partner choice, which may help reduce exposure to domestic violence Oreopoulos (2006).

I exploit the quasi-random variation in schooling induced by a school reform in India, the Right to Education Act (RTE), 2009 to causally estimate the effect of ed-

³https://www.indiacode.nic.in/bitstream/123456789/15436/1/protection_of_women_from_domestic_violence_act%202005.pdf

⁴[https://prsindia.org/files/bills_acts/acts_parliament/2013/criminal-law-\(a\)-act-2013.pdf](https://prsindia.org/files/bills_acts/acts_parliament/2013/criminal-law-(a)-act-2013.pdf)

ucation on violence and related outcomes. The key feature of the reform was that it provides free and compulsory elementary education to kids between the age group of 6 to 14 years students. I use the National Family Health Survey (NFHS)-Round 4 and 5 conducted in 2015-16 and 2019-21 which in addition to the standard survey has a domestic violence module that asks eligible women if they have experienced physical, sexual, or emotional abuse from their current or previous partner. The effect of the reform's implementation is distributed across many birth cohorts and its timing and degree differed across states. Given the continuity in the transition from untreated birth cohorts to treated cohorts in different states, I use an instrumental variable difference-in-difference approach to evaluate the impact of schooling reform as an exogenous variation in education.

This work is a novel attempt to study the Right to Education policy, 2009 thereby complementing the existing literature on the causal connection between female education and intimate partner violence in a developing country context. I first estimate the effect of the Right to Education Act, 2009 on schooling outcomes. Second I will test whether the exogenous increase in schooling has any impact on physical, emotional, and sexual violence, and controlling behavior faced by women. Then, look at how the marriage market outcomes have changed as a result of the reform. Finally, I explore the potential channels through which education and violence might be linked. This research differentiates itself from prior studies in two ways: First, instead of relying on simple correlation like many previous studies, this study uses a plausible exogenous variation in schooling. Second, this study focuses on the effects of compulsory schooling reform on intimate partner violence (IPV) by examining the changes in fe-

male education and partner characteristics whereas previous literature has primarily investigated the causal links between IPV and factors such as labor force participation, income, and assets.

To estimate the education effects on spousal violence measures, I begin my analysis first by testing whether the reform had a positive effect on the level of education in our sample. The result from the first stage estimate shows that the reform increased female schooling by approximately .67 years. Furthermore, second-stage estimates reveal that reform had a substantial impact on sexual and emotional violence experienced by women. However, I find no evidence of reform's effect on other indicators of partner violence. Second-stage estimates demonstrate that women affected by the reform are 3 pp less likely to encounter sexual violence and 7 pp less likely to face emotional violence.

Finally, I examined the potential pathways driving the reduction in the prevalence of various forms of domestic violence. First, I find no evidence indicating that the reform had any significant impact on women's labor market outcomes and attitudes toward domestic violence. However, my findings reveal that an increase in the number of years of schooling had a positive effect on women's attitudes towards domestic violence. Furthermore, an increase in a woman's education could also lead to increased selectivity when finding a partner. These assortative matches may affect violence outcomes through spousal characteristics. I found that the reform led to a change in the age and education between a woman and her husband.

2 Literature review

There exists vast literature studying the causes and consequences of domestic violence. In series of studies, economists have studied the determinants of domestic violence from the perspective of household bargaining models. Using data from the southern Indian state of Kerala, [Agarwal and Panda \(2007\)](#), examined how women's property ownership affects partner violence. They find evidence that ownership of immovable property by women reduces the risk of wife beating, which they attribute to the higher bargaining power of these women. In a randomized control setup, [Haushofer et al. \(2019\)](#) show that unconditional cash transfers in Kenya reduce the occurrence of domestic violence independently of whether the husband or the wife receives the transfer. Similarly, studying families in Brazil and leveraging data from mass layoffs, [Bhalotra et al. \(2021\)](#) estimate that both male and female job loss leads to a large and persistent increase in domestic violence. [Heath et al. \(2020\)](#) show that a national cash transfer program in Mali that primarily targets men decreases domestic violence. Using exogenous changes in labor demand in female-intensive industries, [Aizer \(2010\)](#) shows that a decline in the male-female wage gap reduces violence against women. Her finding is in line with the household bargaining model that predicts a relative improvement in female income reduces her exposure to spousal violence by increasing her bargaining power. Exploiting the phased rollout of MGNREGS across districts in India and employing a difference-in-differences strategy, [Sarma \(2022\)](#) showed that the introduction of the MGNREGS reduced officially reported domestic violence crimes.

[Amaral \(2017\)](#) investigates the effect of improved inheritance rights on violence against women by using an exogenous change in the inheritance legislation in India. The change in inheritance law increased women's property rights and economically empowered them. Their results suggest that the women affected by the law are less likely to face spousal violence. [La Mattina \(2017\)](#) explored the causal effect of civil conflict on domestic violence in Rwanda. The conflict decreased the number of surviving men relative to women giving them bargaining power in the marriage market. Their results show that genocide intensity increased the probability of intimate partner violence. [Stevenson and Wolfers \(2006\)](#) find that the adoption of unilateral divorce laws in the United States reduced female suicide rates, the incidence of domestic violence, and the rates of females murdered by their partners. [Roychowdhury and Dhamija \(2021\)](#) examined the causal effect of women's age at marriage on the prevalence of domestic violence. They utilize variation in age at menarche to obtain exogenous variation in women's age at marriage. Their results indicate a delay in women's marriage causes a significant decline in physical violence but has no impact on sexual or emotional violence. [Chatterjee and Poddar \(2020\)](#) analyzed a multidimensional women empowerment program from India to estimate its effects on intimate partner violence. Using a difference-in-difference design, they found evidence of lesser emotional and physical violence for women potentially exposed to the program. Overall, women face lesser intimate partner violence and controlling behavior because of the empowerment intervention.

However, studies also report the existence of the "male backlash", whereby an increase in women's bargaining power leads to an increase in domestic violence

to reassert male dominance. Using four waves of the Bangladesh Demographic and Health Survey (DHS), [Kagy \(2014\)](#) estimated the impact of increased employment opportunities for women using a difference-in-difference specification. She found that increases in female empowerment after trade liberalization are met with an increased likelihood of domestic violence. Rose and Rosemary examined the effects of employment on women's risks of spousal violence. The results show that woman's employment lowers risks of spousal abuse when their male partners are also employed but substantially increases risks when their male partners are not employed. In the Indian context, [Eswaran and Malhotra \(2011\)](#), using NFHS 2nd round data and employing the instrumental variable model, showed that domestic violence could drastically reduce women's autonomy. This is consistent with a non-cooperative model in which husbands use domestic violence to undermine their wives' bargaining position.

A related body of empirical work documents that men use violence as an instrument to extract resources from wives to increase their consumption. [Bloch and Rao \(2002\)](#) employing probit-IV model on survey data, showed that husbands are more likely to beat their wives when the wife's family is rich because there are more resources to extract, and the returns are greater. Using the deviation of the price of gold at the time of marriage as an exogenous variation, [Menon \(2020\)](#) examined its effect on domestic violence. Findings reveal that an unusually higher price of gold at the time of marriage increases the likelihood of domestic violence. [Calvi and Keskar \(2021\)](#) exploits the Dowry Prohibition Amendment Act in India to estimate its effect on dowry payment and violence. Following the amendment's introduction, they found a decrease in dowry payments but an increase in domestic violence.

Turning to the consequence of domestic violence, [Ramos \(2016\)](#) in a randomized control setting, found that domestic violence impedes female labor productivity in Ecuador, while [Lewbel and Pendakur \(2019\)](#) argue that domestic violence in Bangladeshi families results in decreased consumption efficiency and redistribution of household resources toward men. [Currie et al. \(2022\)](#) utilized novel data that combines crime reports with mother's addresses from birth records to investigate the impact of violence experienced during pregnancy on infant health. The study found that prenatal assault increases rates of extremely low birth weight. [Rawlings and Siddique \(2014\)](#) analyzed the relationship between violence toward the mother and child mortality. They uncovered that children of physically abused mothers have increased likelihood of death within 30 days, one year, and five years after birth.

This research relates to the extended literature on the causal effects of the level of schooling on various outcomes such as returns to education in the labor market ([Angrist and Keueger, 1991](#); [Oreopoulos, 2006](#)), health outcomes ([Lleras-Muney, 2005](#); [Clark and Royer, 2013](#)), fertility behavior ([Black et al., 2008](#); [McCrary and Royer, 2011](#); [Breierova and Duflo, 2004](#)). Only a handful of studies have explored the impact of increased schooling on intimate partner violence and most of them exploited compulsory schooling law in different countries. [Erten and Keskin \(2018\)](#) exploit change in the compulsory schooling law in Turkey to examine the impact of education on domestic violence. The regression discontinuity design estimate on DHS data collected in 2008 shows that the reform increased women's education and improved their labor market

outcomes, with a stronger effect for rural women. Further, they provided evidence that increased education among rural women led to an increase in self-reported psychological violence and financial control behavior. Using DHS data from 2008 and 2014 and employing the Regression discontinuity design, [Akyol and Kirdar \(2020\)](#) showed that the policy reduces physical violence against rural women and is protective against IPV for women overall. Further, they showed that the policy effects are realized through changing partner characteristics and increased schooling for women. Their findings are contradictory to those of [Erten and Keskin \(2018\)](#) which they attribute to the misclassification of two key variables: childhood rural areas and women's employment status by the former. [Zhou et al. \(2021\)](#) used China's Compulsory Schooling Law (CSL) reform as a natural experiment to investigate the impact of education on domestic abuse against women by their spouses. They show that education reduces the probability of women experiencing both physical and sexual abuse from their spouses. Using an instrumental variables approach, the LATE estimate suggests that increased schooling lowers the probability of women being beaten or experiencing forced sexual intercourse.

In the Indian context, studies have looked at the effect of an increase in women's schooling induced by educational reforms or policies on different outcomes. Using a fuzzy regression discontinuity design, ([Sunder, 2018](#)) examined the causal impact of the DPEP (program) launched in India in 1994 on direct beneficiaries and their children. The Results demonstrate that the DPEP increased school access, enrollment, literacy, and years of education for both male and female beneficiaries. He further provides evidence of the Intergenerational effects of DPEP. Children of mothers who were

DPEP beneficiaries had higher test scores in math, vernacular, and English. The study provides evidence that these effects were likely mediated through increased mothers' bargaining power, investments in children's education, and better health behaviors. [Agarwal et al. \(2021\)](#) investigated the impact of female education on age at marriage. They also utilized the District Primary Education Program (DPEP) as a natural experiment and the 2015-16 Demographic Health Survey (DHS) data to estimate education's impact on marriage age. In contrast to the existing literature, they provided evidence for a negative relationship between education and age at marriage and suggest that educated women are cleared from the marriage market faster than less educated women.

[Shah and Steinberg \(2019\)](#) investigated the national trend in educational data around the time of policy implementation. They found that school-going increased after the passage of the law and school infrastructure improved both before and after the policy. They also found that the number of students who have to repeat a grade falls drastically after the policy is enacted, in line with the official provisions of the law. [Agarwal \(2021\)](#) presented a similar finding. They observed that automatic promotion reduced dropout rates for children in the lower secondary age. However, they showed that policy had a negative effect on learning outcomes. Using three waves of National Sample Survey Organisation's Survey of Education data, [Bhattacharya](#) examined completion rates of children from rural low-income families after the passage of the RTE Act. Her finding indicates that there has been a 6% increase in school completion rates of children from low-income backgrounds in the post-RTE period relative to the children from high-income backgrounds. My study is the first to exploit Right to education act, 2009 as a natural experiment to investigate its effect on intimate partner violence.

3 Background and Conceptual framework

3.1 Right to Education Act, 2009

India's Right to Education Act (RTE-2009) builds upon previous efforts targeting education. District Primary Education Program in 1994 and the National Policies on Education (NPE, 1986, 1992) were country's first major initiatives to universalize primary education. These were later followed by the Mid-Day-Meal scheme (MDM-1995) and the Sarva Shiksha Abhiyan (SSA-2001), which have been credited with increasing school attendance. However, despite these efforts, there remain significant disparities in children's health and education outcomes across India (Kingdon (2007); WHO, 2018⁵). RTE-2009 aims to address these inequalities by mandating compulsory schooling, improving school infrastructure, and increasing access to education for disadvantaged groups. The Act also aims to build on the progress made by prior programs and policies by increasing children's exposure to them.

In 2009, Government of India passed the Right to Education Act (RTE) to provide free and compulsory education to all children of the age of six to fourteen years⁶. The act came into effect in the whole of India except the State of Jammu and Kashmir. Following this, state governments were required to implement the Act by passing it in their own state legislatures. However, not all states passed the Act at the same time, and the last states implemented it in 2012, three years after its enactment. Jammu and Kashmir was exempted from the RTE Act until October 2019, when the status was with-

⁵<https://www.who.int/docs/default-source/globalreportspdf/files/who-results-report-1819-clean.pdf>

⁶<https://righttoeducation.in/sites/default/files/Right%20of%20Children%20to%20Free%20and%20Compulsory%20>

drawn with the abrogation of Article 370, bringing the newly-formed Union Territory under the purview of the RTE. The Act was finally implemented in the state in May 2020 (For details, see Table 1).

As per the act it is obligatory for the Government to provide free and compulsory elementary education to each and every child, in a neighborhood school up to class 8 in India. Free education in the act implies that no child is liable to pay fees or any other charges that may prevent him or her from pursuing and completing elementary education which also includes the provisions of textbooks, uniforms, stationery items and special educational material for children with disabilities in order to reduce the burden of school expenses.

‘Compulsory education’ in the act entails providing and ensuring admission, attendance and completion of elementary education by all children in the 6-14 age group by the government and local authorities. For this purpose, the act mandates to keep a check on all neighbourhoods through regular surveys and identifies children who are eligible for receiving an education but are out of school should be admitted to an age-appropriate class and provided with special training to enable the child to come up to age-appropriate learning level. Educational challenges have been prevalent at both the centre and states for many years in India. The Right to Education Act 2009 maps out roles and responsibilities for the centre, state and all local bodies to rectify gaps in their education system in order to enhance the quality of education in the country.

Table 1: Details of state wise enactment of RTE rules

Serial no.	Country Name	Date of implementation of Act
1	Andaman & Nicobar Islands	1/4/2010
2	Andhra Pradesh	1/4/2010
3	Arunachal Pradesh	3/6/2010
4	Assam	3/11/2011
5	Bihar	1/4/2010
6	Chandigarh	1/4/2010
7	Chattisgarh	15/11/2010
8	Dadra&Nagar Haveli	1/4/2010
9	Daman & Diu	1/4/2010
10	Delhi	23/11/2011
11	Goa	2/8/2012
12	Gujrat	18/2/2012
13	Haryana	3/6/2011
14	Himachal pradesh	5/3/2011
15	Jharkhand	14/5/2011
16	Karnataka	28/4/2012
17	Kerala	6/5/2011
18	Lakhshadweep	1/4/2010
19	Madhya pradesh	26/3/2011
20	Maharashtra	11/10/2011
21	Manipur	21/10/2010
22	Meghalaya	1/8/2011
23	Mizoram	28/3/2011
24	Nagaland	21/3/2011
25	Odisha	18/10/2010
26	Puducherry	27/10/2011
27	Punjab	12/10/2011
28	Rajasthan	30/3/2011
29	Sikkim	11/8/2010
30	Tamilnaidu	12/11/2011
31	Tripura	11/7/2011
32	Uttar pradesh	1/4/2010
33	Uttrakhand	1/4/2010
34	West bengal	16/3/2012

Source: <https://righttoeducation.in/resources/states>

The Right to Education Act lays down norms and standards relating to Pupil-Teacher-Ratios (number of children per teacher), classrooms, separate toilets for girls and boys, drinking water facility, number of school-working days, working hours of teachers, etc. The Act provides for rational deployment of teachers by ensuring that the specified Pupil-Teacher Ratio is maintained in every school with no urban-rural imbalance whatsoever. It also mandates appointing appropriately trained teachers i.e. teachers with the requisite entry and academic qualifications.

All schools covered under the Right to Education Act 2009 are obligated to constitute a School Management Committee comprising of a head teacher, local elected representative, parents, community members etc. The committees have been empowered to monitor the functioning of schools and to prepare a school development plan. School Management Committees (SMCs) play a crucial role in strengthening participatory democracy and governance in elementary education. The Act mandates for all private schools to reserve 25 percent of their seats for children belonging to socially disadvantaged and economically weaker sections to boost social inclusion. It is backed by a Grievance Redressal (GR) mechanism that allows people to take action against non-compliance of provisions of the Right to Education Act 2009.

There is a special provision in the act that mandates no child can be held back or expelled from school till Class 8. Coined as a 'No-detention policy', this was considered an important feature of the act which was expected to remove the fear of failure and reduce dropout rates in elementary education. Prior to the implementation of RTE in 2009, retention was a decision taken by the classroom teacher. The decision to pass

or fail depends on a teacher's analysis of academic performance, attendance and other behavioural traits of the students which was very subjective. Instead of failing Right to Education Act introduced the Continuous Comprehensive Evaluation (CCE) system in 2009 to ensure grade-appropriate learning outcomes in schools. Recently government of India scrapped "no detention" policy through an amendment citing a reason that while dropout rates under the earlier system did fall, it led to falling standards of educational achievement⁷.

3.2 Analytical Framework

In addition to establishing the causal link between education and IPV, I also identify potential channels through which the Right to education act, 2009 may affect spousal violence. The most direct link between schooling and the reduction in partner violence is the "knowledge effect". An increase in years of schooling may shift women's perspectives on gender norms and domestic violence, as it expands their access to information, informs them of their rights, and enhances their self-worth. This shift in attitudes, brought about by education, may lead to a decreased tolerance for domestic violence. Second "Autonomy effect": women with higher education earn higher incomes which improves their bargaining power (Aizer, 2010). Thus, schooling reform might lead to economic empowerment which may strengthen their outside options and reduces their risk of experiencing violence. Finally, the "Assortative matching effect": a higher-status woman attracts a husband of similar status who does not resort to domestic violence to reassert control (Amaral, 2017). It can be expected that women affected by the reform, may have married men with more desirable characteristics. This may

⁷https://www.education.gov.in/en/sites/upload_files/mhrd/files/upload_document/rte_2019.pdf

then reduce violence through spousal characteristics.

However, there could be an unintended consequence of increased education that may increase the incidence of domestic violence. "Instrumental violence hypothesis" illustrates that in the presence of asymmetric information related to the husband's gains from marriage, an increase in a women's income could lead to an increase in the husband's threats of violence to extract rent from her ([Bobonis et al., 2013](#); [Bloch and Rao, 2002](#)). Thus, economic empowerment through human capital accumulation may result in the use or threat of violence as a tool to extract resources from their partner. Also, men may respond with violence when women experience greater economic empowerment relative to men and culturally accepted notions of masculinity are threatened, commonly known as the "Male backlash effect" ([Field et al., 2016](#)). Thus, increased schooling and women's better economic conditions may threaten the husband's social status, so he may use violence to ensure his dominance. This research shed light on these mechanisms that relate education to economic empowerment and empowerment within the household.

4 Data

To study the impact of the Right to Education Act, 2009 on intimate partner violence, I rely on the fourth and fifth rounds of the National Family Health Survey (NFHS) conducted in 2015-16 and 2019-2021. NFHS is India's largest household survey for health planning and policy formulation. NFHS-4 and 5 used a stratified two-stage sampling design to get a representative sample from the country. In 2015-16 and 2019-21, it surveyed 568,200 and 610,000 households respectively across 707 districts. In total both rounds contain a sample of approximately 12,000,00+ women of age 15-49 years and 10,000,00+ men of age 15-54 years who were interviewed through separate questionnaires using the computer-assisted personal interviewing method.

The survey collects primarily demographic and health information from women aged 15 to 49 years old, including data on their fertility, weight, marital status, employment status, household decision-making, and socioeconomic characteristics, among other things. The survey also includes information on the level of education, birth month and year and marriage history, which are essential for my analysis.

In addition to the standard survey, the NFHS also includes a domestic violence module that asks eligible women if they have experienced physical, sexual, or emotional abuse from their current or previous partner, which I exploit for my analysis. While all women between the ages of 15 and 49 are asked to participate in the standard survey, only one woman in each household who has been married or partnered is randomly selected to complete the domestic violence module. The NFHS captures four different types of domestic violence: moderate physical violence, severe physical violence, sexual violence, and emotional violence. Further, there are several indicators

Table 2: Summary Statistics for Treated and Untreated (21 to 30 Year-Old) Women

	Untreated	Treated
Years of schooling	7.84 (5.26)	9.59 (4.60)
Woman had a job last week	0.22 (0.41)	0.15 (0.36)
Woman works in agriculture	0.05 (0.21)	0.06 (0.24)
Woman works in non-agriculture	0.16 (0.37)	0.08 (0.27)
Woman works in service	0.06 (0.24)	0.01 (0.12)
Women's age at first birth	20.32 (2.94)	19.31 (1.91)
Partner's years of schooling	8.24 (4.80)	8.71 (4.55)
Gap between male and female schooling within couples	0.99 (4.19)	0.37 (4.18)
Husband's age	31.23 (4.60)	26.66 (3.65)
Gap between husband and wife's ages	4.75 (3.82)	4.65 (3.64)
Husband consume alcoholic beverages	0.29 (0.45)	0.23 (0.42)
Women decides on large household purchases	0.08 (0.27)	0.06 (0.23)
Women usually decides on respondent's health care	0.11 (0.32)	0.08 (0.28)
Women usually decides on decides on visits to family or relatives	0.08 (0.27)	0.06 (0.24)
Women decides what to do with money husband earns	0.07 (0.25)	0.06 (0.24)
Women decides how to spend her earnings	0.18 (0.39)	0.16 (0.36)
Women violence attitude index	0.20 (0.30)	0.15 (0.27)
Experienced severe violence	0.07 (0.26)	0.05 (0.22)
Experienced any sexual violence	0.06 (0.24)	0.05 (0.21)
Experienced any less severe violence	0.27 (0.44)	0.24 (0.43)
Experienced any emotional violence	0.12 (0.33)	0.11 (0.31)
Respondent's father ever beat her mother	0.19 (0.39)	0.18 (0.39)
Respondent is poor	0.37 (0.48)	0.37 (0.48)
Observations	54758	

of social and financial control behavior. All these pieces of information are the key outcome variables in this study.

5 Empirical strategy

A simple OLS regression gives a bias estimate as it does not account for the endogeneity problem of selection bias and reverse causality. There can be unobservable factors that might affect both education and Intimate partner violence. For instance, an educated woman might be characteristically different from those who are less educated. This suggests that the negative relationship between schooling and intimate partner violence could be driven by these omitted characteristics. Moreover, a victim of partner violence may struggle to focus on their studies, miss classes, or drop out altogether which result in a potential reverse causality problem.

5.1 Identification Strategy: The Estimated Likelihood of Being Treated by the 2010 Right to Education Act

To estimate the causal effect of education on intimate partner violence, I rely on the exogenous variation in schooling years provided by the Indian Education Policy 2009. To address potential endogeneity issues, I exploit the variation in program exposure across different birth cohorts to identify the policy's impact on partner violence and employ the instrumental variable (IV) technique. The IV approach is built on the assumption that the likelihood of compulsory education in school is correlated with experiencing the 8 years of minimal education curriculum ($Reform_{isc}$), and the latter only affects

fertility through education (exclusion restriction). The instrument is defined as below:

$$Reform_{isc} = \begin{cases} 1 & \text{if individual } i \text{ in birth cohort } c \text{ is affected by the reform in state } s \\ 0 & \text{otherwise.} \end{cases}$$

In order for the instrument $Reform_{isc}$ dummy to be valid, it needs to satisfy the following assumption. 1) Independence assumption which implies that the instrument is as good as randomly assigned. This means that individuals with different values of the instrument do not systematically differ in any unobserved variables that affect the outcome. The independence assumption can be challenged if there was a correlation between the implementation (timing or intensity) of the reform and pre-reform IPV rates in the states. In most states, reforms took place when marriage violence rates were either decreasing or remaining stagnate, suggesting the timing of the reform was not driven by fertility trends in particular. 2) Monotonicity would violate there are women who are induced by the reform requirements to drop out of school (defiers). This seems counter-intuitive, so it seems plausible to assume there are no defiers. 3) Exclusion restriction - The greatest threat to identification is that the generations most exposed to the law are also most exposed to other social changes. Identification relies on having women from the same generation in both treated and control groups. This comes from two sources- (1) Some individuals born before may be also affected by the reform, the estimate is likely to be biased downwards and will thus be a conservative estimate of the impact of years of education on early fertility. However, this concern

disperses because of the special provision of the reform that does not affect the cohorts around the threshold. The reform is such that if a child above six years of age has never been admitted to school or though admitted, could not complete elementary education, then the child will be admitted to an age-appropriate class and complete elementary education. The reform facilitates a child admitted to an age-appropriate class to be given special training to enable him or her to be at par with other children

(2) staggered implementation of the law across states.

5.2 2SLS Model

This first stage relationship can be described by the following equation:

$$Educ_{isc} = \beta_0 + \beta_1.Reform_{isc} + \beta_2.X_i + \theta_s + \theta_c + \epsilon_{isc}(Firststage)$$

$Reform_{isc}$ is the potential treatment for the education of the individuals when the policy came into force in different states and β_1 is my coefficient of interest which will measure change in education of women as a result of reform. X_i is the vector of some potential control variables. These covariates are either constant over an individual's life or occur prior to schooling decisions being made. θ_s is the state fixed effects, θ_c is the birth cohort fixed effects where cohort c is defined by the interaction of the birth year and birth month of an individual i, and ϵ_{isc} is the unobserved characteristics. I use the above equation as the fixed effects DD estimator with treatments usually occurring

at different times. Some units of observation are treated at a particular time based on the policy enactment in a particular state as the treatment group and untreated units are interpreted as the control group.

Using the first stage estimates, the main empirical specification is defined by the following equation:

$$Y_{isc} = \pi_0 + \pi_1.Educ_{isc} + \pi_2.X_i + \phi_s + \phi_c + \epsilon_{isc}(secondstage2SLS)$$

where i is the individual, Y is the outcome variable (domestic violence measures). X_i is the vector of some potential control variables- religion, district, urban/rural areas, ethnicity. ϕ_s is the state fixed effects, ϕ_c is the birth cohort fixed effects. π_1 here can be expressed as the ratio of reduced form and first-stage coefficients (IV estimator) which measures the average effect of treatment on the compliers (LATE). This implies estimate is given by the difference in outcomes for the groups intended and not intended for treatment divided by the difference in actual treatment for these groups.

6 Result

The main objective of this study is to estimate the impact of RTE on domestic violence. First I produce OLS estimate. Table (3) shows the OLS estimate of education on violence variables. There is a significant fall in women experiencing domestic violence

and all the variables are significant but there might be an endogeneity issue that needs to be dealt with.

Table 3: OLS estimates of Schooling on Domestic violence indicators

VARIABLES	(1) sexual violence	(2) severe violence	(3) less severe violence	(4) emotional violence	(5) Gender attitude index
Years of schooling	-0.0028*** (0.0004)	-0.0057*** (0.0006)	-0.0124*** (0.0010)	-0.0051*** (0.0005)	-0.0065*** (0.0006)
Constant	0.0756*** (0.0024)	0.1227*** (0.0038)	0.3650*** (0.0066)	0.1657*** (0.0036)	0.2422*** (0.0042)
Observations	62,012	62,012	62,012	62,012	61,964
R-squared	0.0491	0.0656	0.111	0.0663	0.149
F stat	58.06	101.1	157	90.59	106.7
Mean of dependent variable	0.0570	0.0846	0.283	0.132	0.199

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

Table 4: Reduced form estimate of Reform on Domestic Violence

VARIABLES	(1) Sexual violence	(2) Severe violence	(3) less severe violence	(4) Emotional violence	(5) Gender attitude index
Years of schooling	-0.0033*** (0.0003)	-0.0064*** (0.0006)	-0.0138*** (0.0009)	-0.0062*** (0.0006)	-0.0073*** (0.0007)
Constant	0.0845*** (0.0021)	0.1301*** (0.0042)	0.3791*** (0.0061)	0.1753*** (0.0040)	0.2748*** (0.0047)
Observations	127,767	127,767	127,767	127,767	127,647
R-squared	0.0201	0.0378	0.0675	0.0274	0.0945
F stat	99.53	97.12	212	100.1	99.78
Mean of dependent variable	0.0633	0.0890	0.291	0.136	0.228

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

In order to address the issue of endogeneity, I provide the reduced form and 2SLS estimates in Table (4) and (5) respectively with reform as an IV that estimates the effect of education on domestic violence variables. Reduced form estimate in table (4) directly estimate the effect of schooling on domestic violence variables. In reduced form estimate, domestic violence variables are found to be significantly reduced. Women who were effected by reform were less likely to face all form of violence for every one year increase in education.

Table (5) and (6) presents the 2SLS-IV estimate which is my main empirical strategy , it can be seen from the table that reform has a significant impact on education such that it increases women's education by .66 years whereas there is weak evidence of its effect on domestic violence. Reform has a substantive effect on sexual and emotional

Table 5: First stage estimates of Reform on schooling

VARIABLES	(1) Schooling	(2) Schooling with fixed effect	(3) Schooling with fixed effect and controls
Reform	0.941** (0.394)	0.245 (0.455)	0.671*** (0.230)
Observations	47044	47044	47044
R-squared	0.002	0.111	0.264
F stat	3.951	3.951	0.477
Mean of dependent variable	0.134	0.134	0.191

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

Table 6: Second stage estimate of Education on Domestic Violence Outcomes

VARIABLES	(1) Sexual violence	(2) Severe violence	(3) Less severe violence	(4) Emotional violence	(5) Gender attitude index
Years of schooling	-0.0356** (0.0125)	-0.0268 (0.0233)	-0.0287 (0.0309)	-0.0598** (0.0271)	-0.0179 (0.0221)
Observations	21,635	21,635	21,635	21,635	21,618
R-squared	0.409	0.108	0.00304	0.506	0.0209
F stat	3.345	1.327	0.863	4.859	0.658
Mean of dependent variable	0.0549	0.0752	0.267	0.126	0.187

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

violence in a favorable direction whereas has no effect on other domestic violence variables. An increase in education results in a reduction of sexual violence among women by 3 pp and gender attitude improves by 7 pp where the reform has no effect on other indicator of domestic violence.

6.1 Robustness test

My 2SLS-IV estimates for intimate partner Violence will be biased if the instrument is capturing a secular trend in women's propensity to attend and improve their education levels. It is plausible that the instrument, reform, might be measuring an unobserved endogenous shift in the demand for education occurring before 2010. To address this issue, I use some state level characteristics that might be associated with the adoption of the RTE law to see if they have any predictive power. In order to do this, I rerun my 2SLS-IV equation while controlling for the state level GDP per capita per capita income

level of a state before the reform interacted with the linear trend in birth cohort. This alternative control allows me to measure indirectly the per capita income level of a state prior to the reform. The rationale behind this approach is that varying state per capita incomes might influence the likelihood of early law adoption, potentially confounding the instrument. Although I lose the significance level but the direction of changes in the magnitude of the coefficients remains similar to the baseline results as shown in above Table.

Table 7: Second stage estimate: Controlling for State Level Pre-Treatment GDP Per-Capita Interacted with Linear Birth Cohort

VARIABLES	(1) sexual violence	(2) severe violence	(3) less severe violence	(4) emotional violence	(5) Gender attitude index
Years of schooling	-0.0343* (0.0220)	-0.0130 (0.0301)	-0.0012 (0.0522)	-0.0847** (0.0364)	-0.0522 (0.0315)
Observations	20,927	20,927	20,927	20,927	20,910
R-squared	-0.382	3.24e-05	0.00375	-1.110	-0.520
F stat	0.667	0.188	0.000556	1.628	2.742
Mean of dependent variable	0.0550	0.0755	0.268	0.126	0.187

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

7 Potential Mechanism

I now examine the potential mechanism underlying the reduced domestic violence experienced by women. In this section I investigate how an increase in women's education those exposed to RTE can lead to decline in women's education on domestic violence against women.

7.1 Labour Market Outcomes and Intrahousehold Decision-making

Increase in a women's relative income increases her outside option. An increase in outside options improves intra-household bargaining power which may reduce the threat

of domestic violence. However, Instrument theory of violence suggests an increase in income due to education may incentive the male partner to use domestic violence as a resource extraction. Hence, i investigate these two channels to explore the effect of the increase in female education on overall employment, employment outside the family activities, income, and assets. Also, improved labor market outcomes may also improve likelihood of violence reporting.

To test the effect of an increase in education on labor force participation, using the Instrumental variable difference-in-difference methodology. Table 7 shows the impact of the program. Although the RTE program does not significantly impact working or participating in the labor market for women, the coefficient suggests a negative effect, as shown in column 1 of Table 8. Also, the results imply that the intrahousehold decision-making power of RTE-treated women did not increase. Furthermore, this finding can be seen in columns 1,2,3 4, and 5 in Table 9, where we observe that all variables are insignificant.

Table 8: Impact of RTE on Women's Labour Outcomes

VARIABLES	(1) Employed	(2) Employed in Non-agriculture	(3) Employed in service
Years of schooling	-0.0133 (0.0199)	0.0053 (0.0186)	0.0042 (0.0096)
Observations	21,635	21,635	21,635
R-squared	0.00246	0.00238	0.0233
F stat	0.446	0.0797	0.195
Mean of dependent variable	0.205	0.102	0.0193

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

Table 9: Impact of RTE on Women's Intrahousehold Bargaining Power

VARIABLES	(1) Her family visits	(2) Own health care	(3) Household purchases	(4) Husband's earning	(5) Own earnings
Years of schooling	-0.0020 (0.0158)	0.0063 (0.0128)	0.0034 (0.0143)	0.0150 (0.0130)	-0.0424 (0.0406)
Observations	16,908	16,953	16,487	15,799	3,822
R-squared	0.000878	-0.0129	-0.00851	-0.0989	-0.271
F stat	0.0162	0.240	0.0561	1.343	1.091
Mean of dependent variable	0.0814	0.0990	0.0725	0.0647	0.181

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

7.2 Gender Role and Domestic violence attitude

Increasing women's education may lead to progressive gender beliefs. Educated women endorsing gender rights can deter domestic violence, promoting positive attitudes. better gender equality beliefs correlate with increased reporting and higher violence opportunity costs. However, educated women may underreport due to societal honor concerns, potentially complicating the relationship between education and gender attitudes. The impact of education on women's gender beliefs remains ambiguous.

Table 10: Impact of RTE on Women's Attitudes towards Domestic Violence

VARIABLES	(1) Refuse sex	(2) Burns food	(3) Going out	(4) Neglect child	(5) Argues
Years of schooling	-0.0155 (0.0268)	-0.0494** (0.0239)	-0.0174 (0.0253)	0.0030 (0.0374)	-0.0138 (0.0286)
Observations	21,525	21,559	21,577	21,562	21,548
R-squared	0.0160	0.332	0.00615	0.00594	0.00144
F stat	0.333	3.951	0.477	0.00626	0.234
Mean of dependent variable	0.108	0.134	0.191	0.274	0.225

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

In table 10, I show the result on various indicator of women's attitudes toward domestic violence. Although most of the indicators of gender role and domestic violence attitude is found to be negative but all of them except burns food is insignificant. Hence there is not much improvement in women's attitude towards domestic violence as a result of RTE.

7.3 Husband/partner's characteristics

Increase in the number of years of schooling of female partners may lead to better matching in the marriage market ([Agarwal et al., 2021](#)). As educated women marry in educated and wealthier households, which can reduce financial stress and decrease

domestic violence post-marriage.

Table 11: Impact of RTE on Husband's Characteristics

VARIABLES	(1) Partner is Employed	(2) Schooling Difference between Partners	(3) Age Difference between Partners
Years of schooling	-0.0287 (0.0319)	-0.6319** (0.2011)	-0.7238** (0.2765)
Observations	21,173	21,547	21,162
R-squared	0.103	0.155	0.764
F stat	0.809	3.432	6.854
Mean of dependent variable	0.794	0.520	4.938

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

Table 12: Impact of RTE on Men's attitude towards domestic violence

VARIABLES	(1) Disrespectful	(2) unfaithful
Years of schooling	-0.0236 (0.0266)	-0.0215 (0.0321)
Observations	21,531	21,518
R-squared	-0.0284	-0.0240
F stat	0.787	0.449
Mean of dependent variable	0.321	0.199

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

Tables 11 and 12 show the impact of RTE on partner characteristic, such as, employment, gender attitudes, , education and age difference. I find that there is no improvement in the husband/partner's gender attitudes towards domestic violence and also labour market outcome of partner is found to be insignificant. However, As can be seen from column 2 and 3 of table (9), Difference in schooling between partners has reduced .66 years and also the .72 years of age difference has between between partners who were exposed to the program.

7.4 Access to Information

Women with access to information have better understanding of gender rights and thus may resist the violence. Also, access to information and knowledge, may help them to report violence to the police which is otherwise a tedious process.

Table 13: Impact of RTE on Access to Information

VARIABLES	(1) Read newspaper	(2) Bank account	(3) Use mobile	(4) Read text
Years of schooling	0.0610** (0.0492)	0.0023 (0.0377)	0.0991*** (0.0335)	0.0527** (0.0224)
Observations	21,635	21,635	21,635	20,447
R-squared	-0.108	0.0102	-0.0548	0.138
F stat	0.0503	0.0129	0.0875	0.323
Mean of dependent variable	0.294	0.788	0.620	0.621

Note: ***, **, and * indicate statistical significance at the 1%, 5% and 10% levels, respectively.

Tables 13 show the results for the impact of RTE on access to information using various indicators. The results indicate that increased years of schooling have a positive and statistically significant association with the likelihood of reading newspapers, using mobile phones, and reading text. However, no significant relationship was found between education and having a bank account. RTE has resulted in increase in reading newspaper by 6%, increased mobile phones usage by 9% and text reading by 5% respectively.

8 Conclusion

This paper's main goal is to provide empirical evidence on the impact of a schooling reform that exogenously increased the average education level on the prevalence of domestic violence in India. I employ an instrumental variable difference-in-difference approach to estimate the causal effects of the education reform on domestic violence measures and related outcomes.

Previous studies mainly focused on basic correlations between education levels and domestic violence risk. However, these correlations are prone to omitted variable bias, as factors like ability, socioeconomic status, and upbringing can influence

both years of schooling and domestic violence risk. My key contribution is assessing the effects of an exogenous increase in education on various aspects of intimate partner violence in a developing-country context characterized by a high prevalence of domestic violence and relatively low levels of women's empowerment.

My analysis reveals that the reform resulted in an average increase of 0.67 additional years of schooling for women, leading to positive effects on some aspects of intimate partner violence, particularly through changes in the partner's characteristics. Women in the treated cohorts were more likely to be employed, especially in non-agricultural sectors, and had higher personal incomes.

Regarding the experience of violence, my findings show a favorable impact on sexual and emotional violence experienced by women. However, I did not find significant evidence of an impact on severe or less severe violence against these women. I also explored whether the reform-induced increase in years of schooling influenced women's attitudes and found a significant impact on attitudes towards domestic violence whereas, no effect on gender attitudes. Additionally, my research uncovered an impact on the quality of partners, including their education and age gap, and employment status which were found to be improved.

Caution in my finding is that my results are based on a short- to medium-term impact of the reform, primarily among women aged 21 to 30 in 2019-21, many of whom had not yet formed long-term partnerships at the time of the survey. Consequently, the long-term effects of the reform might be more substantial than observed in this relatively young age group.

In summary, my findings suggest that the Right to Education Act in India

had significant empowerment effects through assortative matching and changes in attitudes towards domestic violence. However, this assortative matching have translated into empowerment within the household. Our evidence presents a nuanced perspective on the effectiveness of broad expansions of such schooling reform in a context marked by significant limitations on women's rights, including efforts to reduce domestic violence against women.

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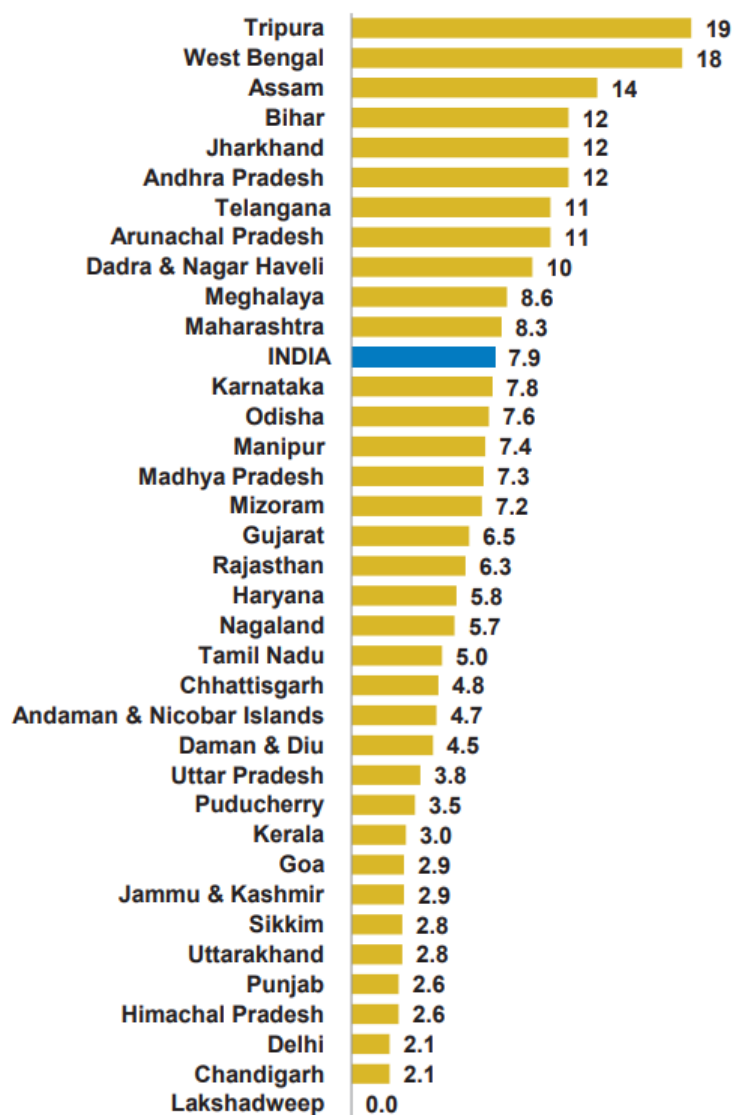
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Figure 1: Spousal violence by state/UT

Percentage of women age 15-19 who have begun childbearing



Source: National Family Health Survey (2015-16)